



PHILOSOPHY OF SCIENCE

Lecture 5: Scientific Realism and its Critiques

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Tuesday, 29 September 2026



OUTLINE

- (1) **Historical background** about realism
- (2) **Scientific realism:** kinds and arguments
- (3) **Van Fraassen's** constructive empiricism
- (4) **Laudan's** pessimistic meta-induction argument

(1) REALISM:
HISTORICAL
BACKGROUND

DEBATES ABOUT THE NATURE OF REALITY

Ontological question:

Does the external world *exist*, independently of us (thoughts and perceptions)?

Metaphysically realist answer: Yes.

- Things/objects exist independently of the mind
- Many philosophers from Plato to the present

Idealist answer: No. The world is mind-dependent

- Berkeley: *Esse est percipi* ('to be is to be perceived')



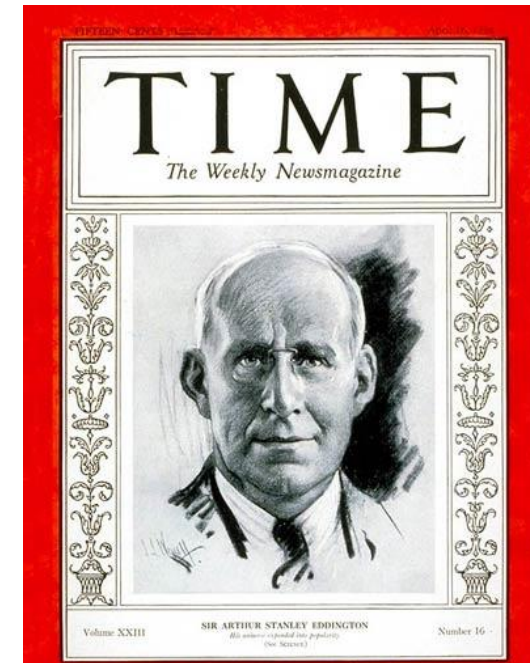
IMMANUEL KANT'S COMPROMISE: NOUMENA VS. PHENOMENA

- **Epistemological question:**
What *knowledge* of the external world can we have?
- There is an external reality, but we cannot have knowledge of it
 - 'Things-in-themselves', *noumena*
- We can only know the appearances (*phenomena*), which are structured by our mind:
 - Space and time: forms of intuition
 - Categories: causality, possibility, etc.



SIR ARTHUR EDDINGTON: 'THE NATURE OF THE PHYSICAL WORLD' (1927)

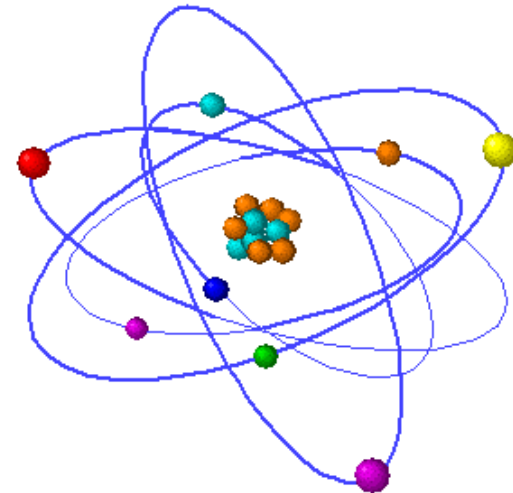
- 'I have settled down to the task of writing these lectures and have drawn up my two chairs to my two tables. [...] One of them has been familiar to me from earliest years. It is a commonplace object of that environment which I call the world. How shall I describe it? It has extension; it is comparatively permanent; it is coloured; above all it is substantial'
- 'Table No. 2 is my scientific table. It is a more recent acquaintance and I do not feel so familiar with it. [...] My scientific table is mostly emptiness. Sparsely scattered in that emptiness are numerous electric charges rushing about with great speed; but their combined bulk amounts to less than a billionth of the bulk of the table itself'



EDDINGTON'S QUESTION

Which of the tables is *real*:

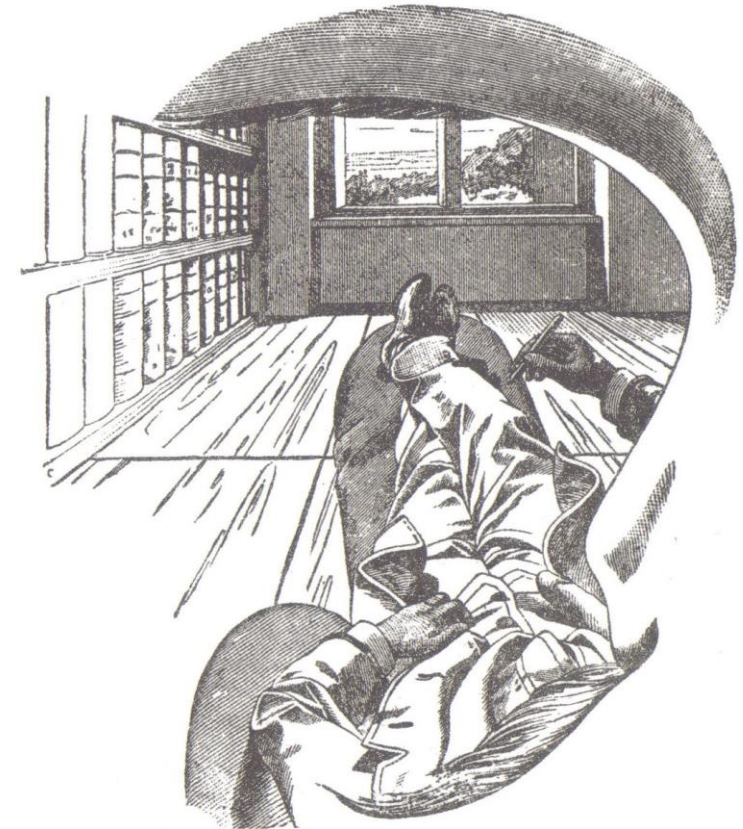
- The table we observe in daily life?
- Or the table as described by science ('almost emptiness')?



ERNST MACH'S RADICAL EMPIRICISM

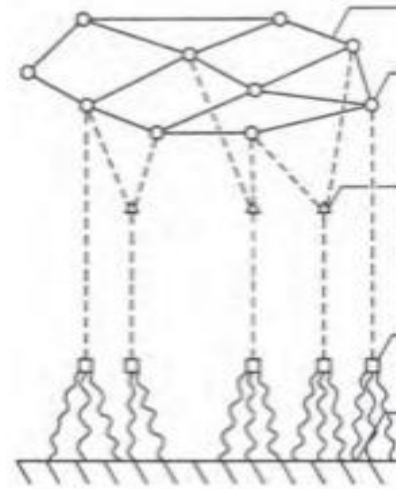
Mach's (1838-1916) **radical empiricism**:

- Claims about unobservable reality are **metaphysical**, cannot have a truth-value.
- Theories are merely economical descriptions of our **sensory experiences**
 - Instrumentalist conception of scientific theories



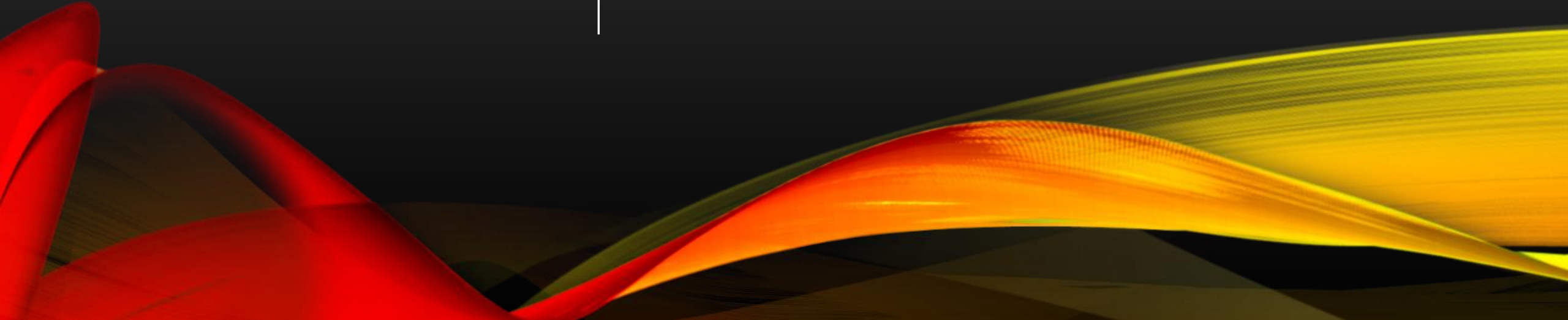
LOGICAL POSITIVISM/LOGICAL EMPIRICISM

- **Early logical positivists:** Carnap and Neurath strict anti-realists, à la Mach:
 - **Verifiability theory of meaning:** claims about the existence of unobservable entities not verifiable, therefore meaningless
 - Sharp theory-observation distinction
 - Theoretical sentences introduced through observational sentences
- **Logical empiricists** gradually shifted to more realist approach:
 - Acceptance of semantics, verification replaced by confirmation.
- **Realist objection to empiricism** (G. Maxwell, 1962):
 - Theory-observation distinction **cannot be sharply drawn.**



Kinds
Arguments

(2) SCIENTIFIC REALISM



WHAT IS SCIENTIFIC REALISM?

- Main contemporary view:
- **Scientific realism:** scientific theories that are **empirically well-confirmed** are true or approximately **true**.
- **Empirical adequacy:** the theory represents/describes/predicts the **observable phenomena** well.
 - Van Fraassen (1980): a theory is empirically adequate if and only if it has at least one model such that all appearances are isomorphic to empirical substructures of that model.



THREE LEVELS/KINDS OF SCIENTIFIC REALISM

- **Semantic realism:** interpret scientific theories literally.
 - Scientific theories aim to tell us a “literal story” about how the world works.
 - Contrast: logical empiricist construal of theoretical terms.
- **Epistemic realism:** one is justified in accepting scientific theories as true.
- **Metaphysical realism:** the theory is *true* or approximately true.
 - The world is like the theory says it is.
- Above definition presupposes all three.

THE NO-MIRACLES ARGUMENT

- **Realism is the only philosophy that does not make the success of science a *miracle*** (Putnam)
 - If scientific theories are not truth-tracking, it is hard to see how they could make successful predictions of **experiments never performed before**.
- Predictions of top quark and Higgs boson were based on theoretical considerations.
 - These theories **predict unobservable entities**:
 - If the theory were false, it is hard to see how these predictions are not incredible coincidences (“miracles”).



INFERENCE TO THE BEST EXPLANATION (IBE)

- No-miracles argument is a case of IBE:
- What is the best explanation of the fact that scientific theories can **predict successfully** the outcomes of experiments never performed before?
- **Approximate truth** of theory is the best explanation of its success:
 - Scientific theories that achieve a certain level of success in prediction and experimental testing are (probably) approximately true.

INFERENCE TO THE BEST EXPLANATION

- Model of reasoning used called IBE or *abduction*
- Comparison between induction and abduction:
 - **Both are ampliative**, i.e. conclusion goes beyond what is (logically) contained in the premises. Both are *fallible*.
 - **Induction**: appeals to observed frequencies/statistics.
 - **Abduction**: also appeals to **explanatory considerations**.
- IBE can be taken to have the following form:
 - Given evidence and candidate explanations ,..., of , infer the truth of that which **best explains** .
- **Approximate truth** of a theory is the best explanation of its empirical **success**
- Anti-realists often do not accept explanations as aims of science.



(3) VAN FRAASSEN'S CONSTRUCTIVE EMPIRICISM

VAN FRAASSEN'S CONSTRUCTIVE EMPIRICISM

Bas van Fraassen's constructive empiricism is a sophisticated alternative to scientific realism.



BAS VAN FRAASSEN AGAINST SCIENTIFIC REALISM

- Scientific Realism:
 - 'Science aims to give us, in its theories, a **literally true** story of what the world is like; and **acceptance** of a scientific theory involves the **belief** that it is true'
- **Constructive Empiricism:**
 - 'Science aims to give us theories which are **empirically adequate**; and **acceptance** of a theory involves as **belief** only that it is empirically adequate'
- **Empirical adequacy:** theory T is empirically adequate: what T says about the *observable* is true.
 - Disagreement with scientific realist is about **unobservable entities**
- Van Fraassen is **semantic realist**: science is a story about the world
 - Epistemic and metaphysical anti-realist.
 - Whether we believe what science says about *unobservable* entities is **up to us**.

VAN FRAASSEN'S CRITIQUE OF SCIENTIFIC REALISM

- Realist arguments that he attacks:
 1. No theory-observation distinction (§2).
 2. Inference to the Best Explanation (§3).
 - *Psychological hypothesis*: we “follow” the IBE rule.
 - Rival hypothesis: we are always willing to believe that the theory which best explains the evidence is **empirically adequate**.
 - Constructive empiricism is a **viable alternative**.
 3. The demand for explanation (§4 & 6).
 4. The Ultimate Argument (§7).

I will only discuss the first argument.

THEORY AND OBSERVATION

Realist argument (G. Maxwell, 1962):

- The theory-observation distinction cannot be sharply drawn.

Van Fraassen (against the logical positivists):

- Take theories *literally* in all respects.
- Theory-observation distinction is **category mistake**:
 - Language = theoretical / non-theoretical
 - Objects = observable / unobservable
- 'Observable' is a **vague, but usable predicate**.





(4) LAUDAN'S PESSIMISTIC META-INDUCTION ARGUMENT

THE PESSIMISTIC META-INDUCTION ARGUMENT

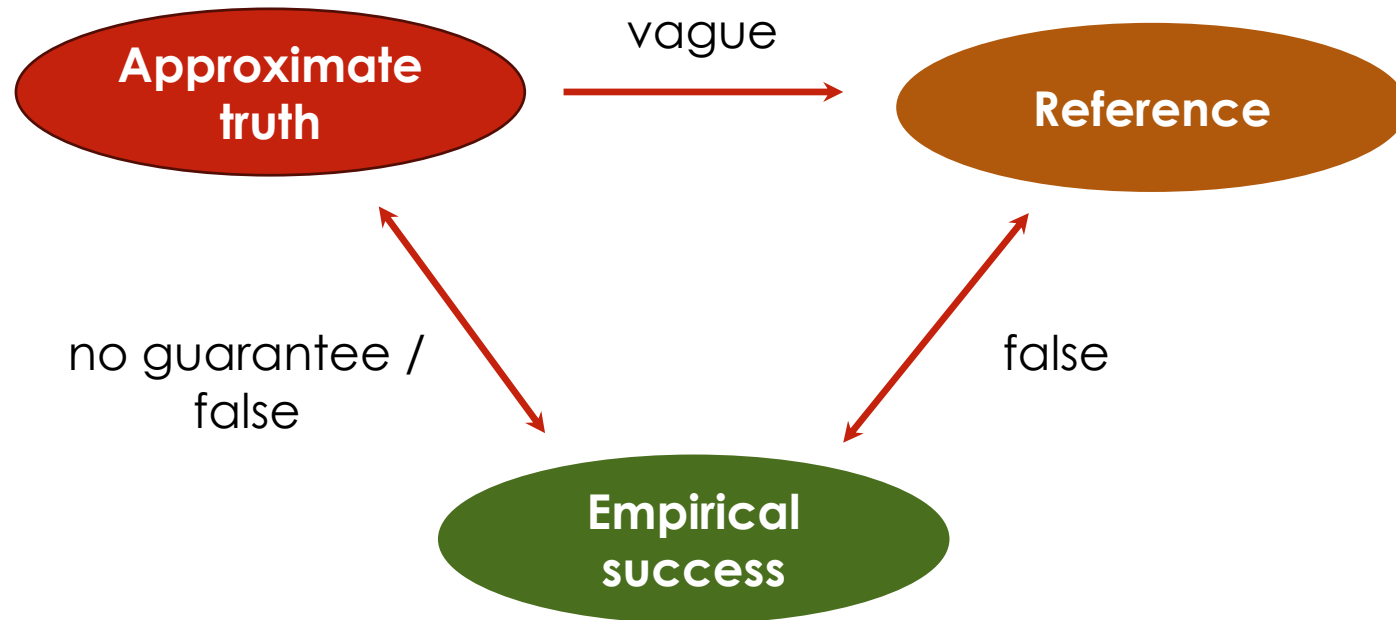
- **History of science:** many theories that we now consider false were once empirically successful
- These theories assumed that terms like 'aether', 'phlogiston', etc. referred to some entities.
- But we do not now take these terms to refer to anything, i.e. these theories assume entities that we now consider to be non-existent.
 - The words for these entities 'do not refer', i.e. there is nothing in the world that corresponds to 'aether' or 'phlogiston'.
- **Induction:** presently successful theories will (may well) turn out to be false too.
 - We are not justified in believing that theories are true.
- There is no reason to suppose that the entities, properties, and processes postulated by our **best theories** are real.

LAUDAN'S EXAMPLES

- crystalline spheres of ancient and medieval astronomy
- humoral theory of medicine
- effluvial theory of static electricity
- catastrophist geology, with its commitment to a universal deluge
- phlogiston theory of chemistry
- caloric theory of heat
- vibratory theory of heat
- vital force theories of physiology
- electromagnetic aether
- optical aether
- theory of circular inertia
- theories of spontaneous generation

'This list, which could be extended *ad nauseam*, involves in every case a theory which was once successful and well confirmed, but which contained central terms which (we now believe) were non-referring.'

CRITIQUE OF CONVERGENT REALISM: TRUTH, REFERENCE AND SUCCESS



Three elements in **convergent realism**:

1. **Approximate truth**: mature theories are closer to the truth
2. **Reference**: mature theories refer and preserve reference
3. **Empirical success**: new theories explain why old theories were successful

THREE RELATED PROBLEMS (LAUDAN, 1981)

1. **There is no good definition of 'approximate truth'.**

- Few of the writers of whom I am aware have defined what it means for a statement or theory to be 'approximately true'
- Old tectonic plate theory, 'tectonic plates' referred, but theory unsuccessful.

2. **Lack of justification for the connection between 'empirical success' and reference assumed by realism.**

- Since approximate truth is vague, we only have reference.
- If the only contact point between empirical success and reference is provided through approximate truth, then the link between reference and success is **extremely tenuous**.

3. **Central terms of theories that were once successful do not refer: hence no approximate truth.**

- 'Aether', 'phlogiston'.

SAATSI (2005) ON PMI

- Two possible conclusions for Laudan's PMI i.e. interpretations of what the induction is about:
 1. **Induction on the past:** current theories are likely to be **false** too.
 2. **Timeless argument that undermines NMA:** attack of realist's argument for optimism. **Success is not a reliable indicator of truth.**
- **NMA:** the best explanation of the success of our current theories is that they are true.

SUMMARY

- Many arguments for and against scientific realism:

For:

1. **No-miracles argument.**
2. **Inference to the best explanation.**

Against:

- **Constructive empiricism** as a viable alternative.
- **Pessimistic meta-induction argument:**
 1. No notion of 'approximate truth'.
 2. No connection between empirical success, reference and approximate truth.
- The debate continues! See next week.